Faycal Kilali

root@faycalkilali.com | 514-886-6179 | github.com/faycalki | linkedin.com/in/faycal-kilali | faycalkilali.com

Education

Université de Montréal – M.Sc. Computer Science (with Internship) – Montreal, QC 2026 – 2028

- Concentrating in AI/ML through **MILA (Quebec AI Institute)**-affiliated coursework at UdeM.
- **UdeM Exemption Scholarship:** $11,004 CAD/year (2026–2028)

Mount Allison University – B.Sc. (Hons) Computer Science and Mathematics – Sackville, NB 2022 – 2026

- **Founder & President**, AI & Robotics Society 2023–2026
  - Grew to **62 members**; recruited **3 faculty advisors** and secured hardware resources & funding for projects
  - Organized project plans and coordinated members around hands-on AI & robotics initiatives.
- **Co-Founder & VP**, Mature Students Club — led recruitment at club fairs, engaging **50+** prospective members; organized peer tutoring sessions in CS & mathematics. 2023–2025

Certifications: *Machine Learning with Python* (IBM), *Object-Oriented Design* (UAlberta), *Microsoft Learn AI Skills* (Microsoft)

Experience

Teaching Assistant, Université de Montréal – Montreal, QC Sep 2026 – Present

- Run weekly **Mathematica/Wolfram Language** labs for undergraduate students, teaching functional & symbolic programming: expressions, lists, functions, recursion, iteration, differentiation, integration, and 2D/3D graphics
- Present and solve lab problems live and hold dedicated office hours, reinforcing computational thinking and mathematical programming fluency across the class

Full-Stack Software Developer, AI Object Detection Framework and Educational Tool May 2024 – Sep 2024
Grant-funded entrepreneurial venture supported by Mount Allison University – Sackville, NB | GitHub

- Secured $5,000 in funding through the $4,000 TD Ignite Grant and a $1,000 third-place Pitch Competition award; led development of an AI-powered educational tool.
- Implemented YOLOv10 object detection with a Flask backend, Streamlit web interface, and Tkinter GUI, supporting 243 languages.
- Built image/video detection workflows and a multilingual educational game using translated object names; integrated REST APIs, multi-platform deployment, and technical documentation.

Mathematics and Computer Science Peer Tutor, Mount Allison University – Sackville, NB Mar 2023 – Sep 2024

- Tutored 50+ students across various mathematical subjects effectively, resulting in positive feedback from students

Technical Lead, Greater Dorchester Moving Forward – Dorchester, NB Jan 2023 – August 2023

- Spearheaded technology operations for a nonprofit organization by leading data collection, mapping, and software development–enhancing impact and advancing the mission
- Collaborated with other departments within the organization to integrate technical solutions relevant to the seated board

Selected Projects

Content-Aware Enterprise Document Search & Management System | Java, PostgreSQL (SQL), JDBC, PDFBox, Bash, MVC, Agile, Scrum, DevOps, Figma | GitHub 2025

- **Architected** a decoupled **MVC** system using **Dependency Injection** to orchestrate state management between the front-end and the application's backend, significantly reducing component coupling and enhancing system modularity
- **Designed** a hybrid search algorithm combining PostgreSQL's `tsvector` for optimized metadata retrieval ($O(\log n)$) with a custom Java-based regex engine for deep content analysis of binary PDF blobs, enabling granular pattern matching
- **Orchestrated** the team's **Agile/Scrum** workflow, serving as **Technical Lead** to mentor peers on Git best practices, resolve merge conflicts, and manage technical debt to ensure on-time delivery of prototypes
- **Established** comprehensive **DevOps** and documentation standards, authoring UML architectural diagrams and configuring **CI/CD** pipelines (Gradle) to standardize the build and release lifecycle for the team

Room Scanner & Voice Guidance System | Python, YOLOv10, Raspberry Pi, Bash, Computer Vision| GitHub 2024

- Engineered embedded room scanning system integrating computer vision with custom hardware circuits (GPIO, LED indicators, push-button controls), implementing both hardware and software debouncing techniques
- Developed real-time voice feedback system for object detection with respect to the physical location of the system, providing information/guidance for blind individuals throughout the observable area
- Created comprehensive documentation of the hardware schematics and the software components with automated installation scripts, thereby emphasizing reproducible deployment

Multi-Paradigm Wordle Puzzle Solver AI Agent | Python, Streamlit, Information Theory, AI | GitHub 2024

- Built an agent combining four AI paradigms (Information-Theoretic, Model-Based, Goal-Based, Utility-Based), using entropy and information gain to select guesses and achieve 3–5 guess performance
- Architected Streamlit visualization dashboard displaying real-time AI decision-making with time complexity of $O(W^2L)$, demonstrating advanced algorithm optimization and probabilistic reasoning

Multivariate Regression Framework | Python, PyTorch, NumPy, Machine Learning, AI | GitHub 2024

- Developed machine learning framework from mathematical foundations, implementing gradient descent optimization, L2 regularization (Ridge), and closed-form solutions using PyTorch and NumPy
- Conducted five experiments examining noise sensitivity, training set impact, parameter complexity, and regularization effects, with visualizations demonstrating theoretical ML concepts in practice

Dreamy Pastures (Multiplayer Board Game Engine) | Java, UML, OOP, Multiplayer | GitHub 2024

- Collaboratively developed local multiplayer board game supporting up to 4 concurrent players with real-time state synchronization and turn-based game logic, demonstrating OOP Design Patterns, and extensive UML Documentation

RSA Encryption & Decryption System | Python, RSA, Security | GitHub 2021

- Implemented RSA cryptography from scratch without cryptographic libraries, developing custom modules for prime generation, modular arithmetic, modular exponentiation, Euler's totient function, and extended GCD to support file and string encryption.

Open Source Video Games | MBScript, Python, PHP | Floris Evolved | Tainted Paths | Medieval Conquests 2015-2019

- Founded and led development of three major game modifications with **314,000+ active subscribers** and **1.3+ million unique visitors** across platforms, achieving **90-95% positive ratings**
- Architected & developed approximately 100,000 lines of MBScript & Python code for game logic and modular systems
- Managed distributed teams of volunteers across multiple years, whereby they assisted with various aspects of game development including translation, art, testing, and design

Skills

Programming Languages: Python, Java, C, C++, BASH/Shell Scripting, MBScript, Assembly (MIPS, RISC-V), LaTeX, SQL, Wolfram Language
Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Flask, Streamlit, YOLOv10, REST API, JDBC, PDFBox, UML, pandas, scikit-learn, Matplotlib, Seaborn
Tools & Platforms: Git, PostgreSQL, Figma, Unity, Linux/Unix, Windows, AWS, Azure, Raspberry Pi, Arduino, Mathematica, Gradle, Keras, OpenCV, NLTK, Conda, Jupyter, GitHub
Machine Learning Methods: Supervised classification & regression (linear & logistic regression, k-nearest neighbors (k-NN), Bayes classifiers, support vector machines (SVMs), decision trees), kernel methods (Parzen windows), ensemble methods (bagging, boosting), clustering (k-means), density estimation, probabilistic graphical models, feature selection & outlier detection, principal component analysis (PCA), maximum likelihood & Bayesian inference
ML Workflow, Optimization & Evaluation: Problem formulation & success metrics, train/validation/test splitting, data leakage prevention, cross-validation (K-fold, stratified K-fold), model selection & hyperparameter tuning, gradient descent, L2 regularization, bias–variance decomposition, overfitting & underfitting, model evaluation (macro F1), error analysis, model robustness, final model training & testing
Data Science & Statistics: Data collection & quality assessment, data cleaning, exploratory data analysis (EDA), data visualization, feature engineering, missing-value imputation, outlier handling, feature scaling, categorical encoding, data transformations, hypothesis testing (t-tests, Mann–Whitney U test), permutation tests, bootstrap confidence intervals
ML Systems & Deployment: Distributed learning, virtualization, model deployment & inference, performance monitoring, data drift detection, reliability & operational constraints, model retraining & updating
AI Foundations & Applications: Information theory, AI agents (information-theoretic, model-based, goal-based, utility-based, logic-based), algorithmic bias, AI applications in science & robotics
Data Science & AI Coursework: Fundamentals of Machine Learning, Data Science, Artificial Intelligence, Intro Data Science
Mathematics: Vector Calculus, Multivariable Calculus, Advanced Linear Algebra, Mathematical Modeling, Probability and Statistics, Discrete Structures, Real Analysis, Mathematical Optimization, Modern Algebra, Differential Equations, Physics
Computer Science: Software Design, Operating Systems, Advanced Data Structures & Algorithms, Database Systems, Computer Networks, Algorithm Analysis, Systems Programming, Digital Electronics and Digital Signal Processing, Simulation and Modeling, Computer Organization & Architecture, Object-Oriented Design **Software Engineering:** Object-Oriented Programming (OOP), MVC Architecture, System Design, Parallel Processing, Procedural Programming, Logic Programming, Database Programming
Algorithmic Paradigms: Divide and Conquer, Greedy Algorithms, Dynamic Programming, Decrease and Conquer
Domain Expertise: Low-level programming, Game Engines, Security/Cryptography, Server Infrastructure, GIS, Digital Signal Processing, Computer Graphics, Haptics, Computer Networks, VR/AR, GPU, CPU
Embedded Systems & Hardware: Hardware Design, Hardware Schematics, Debouncing, Automation
Methodologies: Agile, Scrum, DevOps
Languages: English (Native), French, Standard Arabic (Professional), Saudi Arabic (Professional), Darija (Native)